SONIC bench test: brief results

23 October 2026 Analysis of existing recordings

Setup. Computer → Audient iD14 → Edifier speaker → Earthworks M50 → Audient microphone input 1. Microphone separation was approximately 10 cm (operator reported). Four saved 96 kHz recordings were analyzed; no new test was performed.

Timing. All 40 timing sweeps were accepted across two stream starts.

Pooled median application roundtrip	**38.307 ms**
Observed range	38.240–38.375 ms
Individual run medians	38.250 / 38.375 ms

This demonstrates short-run repeatability for the tested program and setup. The delay includes computer/interface buffering and the acoustic chain; it does not establish calibrated submillisecond accuracy or isolated speaker delay.

SONIC sounds. Two 84-second blocks used a local reconstruction of the paper's 14 Sheep E frequencies, from 2 to 27 kHz, with 10 ms pips and 2 ms ramps: 16,800 pips total. The original SONIC audio files were not used.

Frequency results. Both microphone recordings contain all 14 commanded frequency components, but their levels are uneven. Relative to 2 kHz, the 17–19.5 kHz groups were approximately 15–22 dB lower; 20.9–25.8 kHz were 30–40 dB lower. These are full-chain relative recordings. Calibrated SPL, speaker-only response and distortion remain unmeasured.

Next. Fix and document the geometry, apply microphone/SPL calibration, test separated tones, and validate timing against an independent electrical reference.

Source: full saved-data analysis report and measurement records.